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Simple layers

Simple Modules are used for various tasks like adapting Tensor methods and providing affine transformations :

- Parameterized Modules :
  - Linear : a linear transformation ;
  - LinearWeightNorm : a weight normalized linear transformation ;

    - SparseLinear : a linear transformation with sparse inputs ;
    - IndexLinear : an alternative linear transformation with for sparse inputs and max normalization ;
    - Bilinear : a bilinear transformation with sparse inputs ;
    - PartialLinear : a linear transformation with sparse inputs with the option of only computing a subset ;
    - Add : adds a bias term to the incoming data ;
    - CAdd : a component-wise addition to the incoming data ;
    - Mul : multiply a single scalar factor to the incoming data ;
    - CMul : a component-wise multiplication to the incoming data ;
    - Euclidean : the euclidean distance of the input to `k` mean centers ;
    - WeightedEuclidean : similar to Euclidean, but additionally learns a diagonal covariance matrix ;
    - Cosine : the cosine similarity of the input to `k` mean centers ;
    - Kmeans : Kmeans clustering layer;
- Modules that adapt basic Tensor methods :
    - Copy : a copy of the input with type casting ;
    - Narrow : a narrow operation over a given dimension ;
    - Replicate : repeats input `n` times along its first dimension ;
    - Reshape : a reshape of the inputs ;
    - View : a view of the inputs ;
    - Contiguous : contiguous of the inputs ;
    - Select : a select over a given dimension ;
    - MaskedSelect : a masked select module performs the torch.maskedSelect operation ;
    - Index : a index over a given dimension ;
    - Squeeze : squeezes the input;
    - Unsqueeze : unsqueeze the input, i.e., insert singleton dimension;
    - Transpose : transposes the input ;
- Modules that adapt mathematical Tensor methods :
    - AddConstant : adding a constant ;

- MulConstant : multiplying a constant ;
- Max : a max operation over a given dimension ;
- Min : a min operation over a given dimension ;
- Mean : a mean operation over a given dimension ;
- Sum : a sum operation over a given dimension ;
- Exp : an element-wise exp operation ;
- Log : an element-wise log operation ;
- Abs : an element-wise abs operation ;
- Power : an element-wise pow operation ;
- Square : an element-wise square operation ;
- Sqrt : an element-wise sqrt operation ;
- Clamp : an element-wise clamp operation ;
- Normalize : normalizes the input to have unit `L_p` norm ;
- MM : matrix-matrix multiplication (also supports batches of matrices) ;

- Miscellaneous Modules :
  - BatchNormalization : mean/std normalization over the mini-batch inputs (with an optional affine transform) ;
  - PixelShuffle : Rearranges elements in a tensor of shape `[C*r, H, W]` to a tensor of shape `[C, H*r, W*r]` ;
  - Identity : forward input as-is to output (useful with ParallelTable) ;
  - Dropout : masks parts of the `input` using binary samples from a bernoulli distribution ;
  - SpatialDropout : same as Dropout but for spatial inputs where adjacent pixels are strongly correlated ;
  - VolumetricDropout : same as Dropout but for volumetric inputs where adjacent voxels are strongly correlated ;
  - Padding : adds padding to a dimension ;
  - L1Penalty : adds an L1 penalty to an input (for sparsity) ;
  - GradientReversal : reverses the gradient (to maximize an objective function) ;
  - GPU : decorates a module so that it can be executed on a specific GPU device.
  - TemporalDynamicKMaxPooling : selects the k highest values in a sequence. k can be calculated based on sequence length ;
  - Constant : outputs a constant value given an input (which is ignored);

- WhiteNoise : adds isotropic Gaussian noise to the signal when in training mode;
- OneHot : transforms a tensor of indices into one-hot encoding;
- PrintSize : prints the size of `input` and `gradOutput` (useful for debugging);
- ZeroGrad : forwards the `input` as-is, yet zeros the `gradInput`;
- Collapse : just like `nn.View(-1)`;
- Convert : convert between different tensor types or shapes;

Linear

```
module = nn.Linear(inputDimension, outputDimension, [bias = true])
```

Applies a linear transformation to the incoming data, i.e. `y = Ax + b`. The `input` tensor given in `forward(input)` must be either a vector (1D tensor) or matrix (2D tensor). If the input is a matrix, then each row is assumed to be an input sample of given batch. The layer can be used without bias by setting `bias = false`.

You can create a layer in the following way:

```
module = nn.Linear(10, 5)  -- 10 inputs, 5 outputs
```

Usually this would be added to a network of some kind, e.g.:

```
mlp = nn.Sequential()
mlp:add(module)
```

The weights and biases (*A* and *b*) can be viewed with:

```
print(module.weight)
print(module.bias)
```

The gradients for these weights can be seen with:

```
print(module.gradWeight)
print(module.gradBias)
```

As usual with `nn` modules, applying the linear transformation is performed with:

```
x = torch.Tensor(10) -- 10 inputs
y = module:forward(x)
```

LinearWeightNorm

```
module = nn.LinearWeightNorm(inputDimension, outputDimension, [bias = true])
```

LinearWeightNorm implements the reparametrization presented in Weight Normalization, which decouples the length of neural network weight vectors from their direction. The weight vector `w` is determined instead by parameters `g` and `v` such that `w = g * v / ||v||`, where `||v||` is the euclidean norm of vector `v`. In all other respects this layer behaves like `nn.Linear`.

To convert between `nn.Linear` and `nn.LinearWeightNorm` you can use the `nn.LinearWeightNorm.fromLinear(linearModule)` and `weightNormModule:toLinear()` functions.

Other layer types can make use of weight normalization through the nn.WeightNorm container.

SparseLinear

```
module = nn.SparseLinear(inputDimension, outputDimension)
```

Applies a linear transformation to the incoming sparse data, i.e. `y = Ax + b`. The `input` tensor given in `forward(input)` must be a sparse vector represented as 2D tensor of the form torch.Tensor(N, 2) where the pairs represent indices and values. The SparseLinear

layer is useful when the number of input dimensions is very large and the input data is sparse.

You can create a sparse linear layer in the following way:

```
module = nn.SparseLinear(10000, 2)   -- 10000 inputs, 2 outputs
```

The sparse linear module may be used as part of a larger network, and apart from the form of the input, SparseLinear operates in exactly the same way as the Linear layer.

A sparse input vector may be created as so...

```
x = torch.Tensor({ {1, 0.1}, {2, 0.3}, {10, 0.3}, {31, 0.2} })

 print(x)

  1.0000   0.1000
  2.0000   0.3000
 10.0000   0.3000
 31.0000   0.2000
[torch.Tensor of dimension 4x2]
```

The first column contains indices, the second column contains values in a a vector where all other elements are zeros. The indices should not exceed the stated dimensions of the input to the layer (10000 in the example).

IndexLinear

```
module = nn.IndexLinear(inputSize, outputSize, doGradInput, keysOffset, weight, bias, normalize)
```

Applies the following transformation to the incoming (optionally) normalized sparse input data: `z = Weight * y + bias`, where

- `y_i = normalize and (x_i *  (1 / x_i_max) + b_i) or x_i`
- `x_i` is the `i'th` feature of the input,

- `b_i` is a per-feature bias,
- `x_i_max` is the maximum absolute value seen so far during training for feature `i`.

The normalization of input features is very useful to avoid explosions during training if sparse input values are really high. It also helps ditinguish between the presence and the absence of a given feature.

Parameters

- `inputSize` is the maximum number of features.
- `outputSize` is the number of output neurons.
- `doGradInput`, if `false` (the default), the gradInput will not be computed.
- `keysOffset` lets you specify input keys are in the `[1+keysOffset, N+keysOffset]` range. (defaults to `0`)
- `weight` and `bias` allow you to create the module with existing weights without using additional memory. When passing `weight` and `bias`, `inputSize` and `outputSize` are inferred from the weights.
- `normalize` will activate the normalization of the input feature values. (`false` by default)

You can create an `IndexLinear` layer the following way:

```
-- 10000 inputs, 2 outputs, no grad input, no offset, no input weight/bias, max-norm on
module = nn.IndexLinear(10000, 2, nil, 0, nil, nil, true)
```

Differences from SparseLinear

- The layout of `weight` is transposed compared to `SparseLinear`. This was done for performance considerations.
- The `gradWeight` that is computed for in-place updates is a sparse representation of the whole gradWeight matrix. Its size changes from one backward pass to another. This was done for performance considerations.
- The input format differs from the SparseLinear input format by accepting keys and values as a table of tensors. This enables `IndexLinear` to have a larger range for keys than `SparseLinear`.

The `input` tensors must be in one of the following formats.

- An array of size 2 containing a batch of `keys` followed by a batch of `values`.

```
x = {
      { torch.LongTensor({ 1, 200 }), torch.LongTensor({ 100, 200, 1000 }) },
      { torch.Tensor({ 1, 0.1 }), torch.Tensor({ 10, 0.5, -0.5 }) }
}
```

- an array of size 3 containing a flattened (pre-concatenated) batch of `keys`, followed by `values`, and `sizes`.

```
-- Equivalent to the input shown above
x = {
      torch.LongTensor({ 1, 200, 100, 200, 1000 }),
      torch.Tensor({ 1, 0.1, 10, .5, -0.5 }),
      torch.LongTensor({ 2, 3 })
}
```

Note: The tensors representing `keys` and `sizes` must always be of type `LongTensor` / `CudaLongTensor`. The values can be either `FloatTensor` or `DoubleTensor` or their cutorch equivalents.

Bilinear

```
module = nn.Bilinear(inputDimension1, inputDimension2, outputDimension, [bias = true])
```

Applies a bilinear transformation to the incoming data, i.e. `\forall k: y_k = x_1 A_k x_2 + b`. The `input` tensor given in `forward(input)` is a table containing both inputs `x_1` and `x_2`, which are tensors of size `N x inputDimension1` and `N x inputDimension2`, respectively. The layer can be trained without biases by setting `bias = false`.

You can create a layer in the following way:

```
 module = nn.Bilinear(10, 5, 3)  -- 10 and 5 inputs, 3 outputs
```

Input data for this layer would look as follows:

```
input = {torch.randn(128, 10), torch.randn(128, 5)}  -- 128 input examples
module:forward(input)
```

PartialLinear

```
module = nn.PartialLinear(inputSize, outputSize, [bias = true])
```

PartialLinear is a Linear layer that allows the user to a set a collection of column indices. When the column indices are set, the layer will behave like a Linear layer that only has those columns. Meanwhile, all parameters are preserved, so resetting the PartialLinear layer will result in a module that behaves just like a regular Linear layer.

This module is useful, for instance, when you want to do forward-backward on only a subset of a Linear layer during training but use the full Linear layer at test time.

You can create a layer in the following way:

```
module = nn.PartialLinear(5, 3)  -- 5 inputs, 3 outputs
```

Input data for this layer would look as follows:

```
input = torch.randn(128, 5)  -- 128 input examples
module:forward(input)
```

One can set the partition of indices to compute using the function `setPartition(indices)` where `indices` is a tensor containing the indices to compute.

```
module = nn.PartialLinear(5, 3)  -- 5 inputs, 3 outputs
module:setPartition(torch.Tensor({2,4})) -- only compute the 2nd and 4th indices out of a total of 5 indices
```

One can reset the partition via the `resetPartition()` function that resets the partition to compute all indices, making it's behaviour equivalent to `nn.Linear`

Dropout

```
module = nn.Dropout(p)
```

During training, `Dropout` masks parts of the `input` using binary samples from a bernoulli distribution. Each `input` element has a probability of `p` of being dropped, i.e having its commensurate output element be zero. This has proven an effective technique for regularization and preventing the co-adaptation of neurons (see Hinton et al. 2012).

Furthermore, the outputs are scaled by a factor of `1/(1-p)` during training. This allows the `input` to be simply forwarded as-is during evaluation.

In this example, we demonstrate how the call to forward samples different `outputs` to dropout (the zeros) given the same `input`:

```
module = nn.Dropout()

> x = torch.Tensor{{1, 2, 3, 4}, {5, 6, 7, 8}}

> module:forward(x)
  2   0   0   8
 10   0  14   0
[torch.DoubleTensor of dimension 2x4]

> module:forward(x)
  0   0   6   0
 10   0   0   0
[torch.DoubleTensor of dimension 2x4]
```

Backward drops out the gradients at the same location:

```
> module:forward(x)
```

```
  0   4   0   0
 10  12   0  16
[torch.DoubleTensor of dimension 2x4]

> module:backward(x, x:clone():fill(1))
 0  2  0  0
 2  2  0  2
[torch.DoubleTensor of dimension 2x4]
```

In both cases the `gradOutput` and `input` are scaled by `1/(1-p)`, which in this case is `2`.

During evaluation, `Dropout` does nothing more than forward the input such that all elements of the input are considered.

```
> module:evaluate()

> module:forward(x)
 1  2  3  4
 5  6  7  8
[torch.DoubleTensor of dimension 2x4]
```

There is also an option for stochastic evaluation which drops the `outputs` just like how it is done during training:

```
module_stochastic_evaluation = nn.Dropout(nil, nil, nil, true)

> module_stochastic_evaluation:evaluate()

> module_stochastic_evaluation:forward(x)
  2   4   6   0
  0  12  14   0
[torch.DoubleTensor of dimension 2x4]
```

We can return to training our model by first calling Module:training():

```
> module:training()
```

```
> return module:forward(x)
  2    4    6    0
  0    0    0   16
[torch.DoubleTensor of dimension 2x4]
```

When used, `Dropout` should normally be applied to the input of parameterized Modules like Linear or SpatialConvolution. A `p` of `0.5` (the default) is usually okay for hidden layers. `Dropout` can sometimes be used successfully on the dataset inputs with a `p` around `0.2`. It sometimes works best following Transfer Modules like ReLU. All this depends a great deal on the dataset so its up to the user to try different combinations.

SpatialDropout

`module` = `nn.SpatialDropout(p)`

This version performs the same function as `nn.Dropout`, however it assumes the 2 right-most dimensions of the input are spatial, performs one Bernoulli trial per output feature when training, and extends this dropout value across the entire feature map.

As described in the paper "Efficient Object Localization Using Convolutional Networks" (http://arxiv.org/abs/1411.4280), if adjacent pixels within feature maps are strongly correlated (as is normally the case in early convolution layers) then iid dropout will not regularize the activations and will otherwise just result in an effective learning rate decrease. In this case, `nn.SpatialDropout` will help promote independence between feature maps and should be used instead.

`nn.SpatialDropout` accepts 3D or 4D inputs. If the input is 3D than a layout of (features x height x width) is assumed and for 4D (batch x features x height x width) is assumed.

VolumetricDropout

`module` = `nn.VolumetricDropout(p)`

This version performs the same function as `nn.Dropout`, however it assumes the 3 right-most dimensions of the input are spatial, performs one Bernoulli trial per output feature when training, and extends this dropout value across the entire feature map.

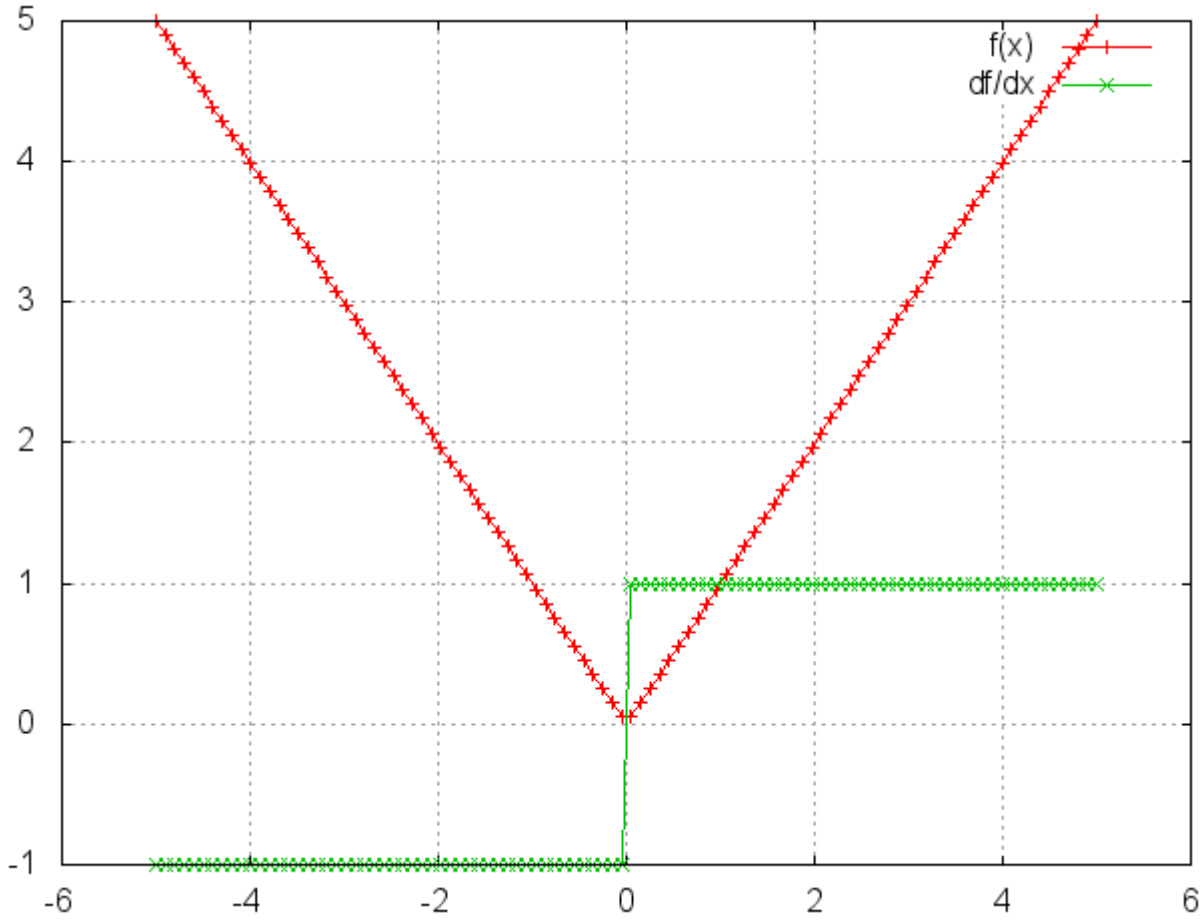
As described in the paper "Efficient Object Localization Using Convolutional Networks" (http://arxiv.org/abs/1411.4280), if adjacent voxels within feature maps are strongly correlated (as is normally the case in early convolution layers) then iid dropout will not regularize the activations and will otherwise just result in an effective learning rate decrease. In this case, `nn.VolumetricDropout` will help promote independence between feature maps and should be used instead.

`nn.VolumetricDropout` accepts 4D or 5D inputs. If the input is 4D than a layout of (features x time x height x width) is assumed and for 5D (batch x features x time x height x width) is assumed.

Abs

```
module = Abs()
```

```
m = nn.Abs()
ii = torch.linspace(-5, 5)
oo = m:forward(ii)
go = torch.ones(100)
gi = m:backward(ii, go)
gnuplot.plot({'f(x)', ii, oo, '+-'}, {'df/dx', ii, gi, '+-'})
gnuplot.grid(true)
```



Add

```
module = nn.Add(inputDimension, scalar)
```

Applies a bias term to the incoming data, i.e. `yi = x_i + b_i`, or if `scalar = true` then uses a single bias term, `yi = x_i + b`. So if `scalar = true` then `inputDimension` value will be disregarded.

Example:

```
y = torch.Tensor(5)
mlp = nn.Sequential()
mlp:add(nn.Add(5))

function gradUpdate(mlp, x, y, criterion, learningRate)
   local pred = mlp:forward(x)
   local err = criterion:forward(pred, y)
   local gradCriterion = criterion:backward(pred, y)
   mlp:zeroGradParameters()
   mlp:backward(x, gradCriterion)
   mlp:updateParameters(learningRate)
   return err
end

for i = 1, 10000 do
   x = torch.rand(5)
   y:copy(x);
   for i = 1, 5 do y[i] = y[i] + i; end
   err = gradUpdate(mlp, x, y, nn.MSECriterion(), 0.01)
end

print(mlp:get(1).bias)
```

gives the output:

```
 1.0000
 2.0000
 3.0000
 4.0000
 5.0000
[torch.Tensor of dimension 5]
```

i.e. the network successfully learns the input `x` has been shifted to produce the output `y`.

CAdd

```
module = nn.CAdd(size)
```

Applies a component-wise addition to the incoming data, i.e. `y_i = x_i + b_i`. Argument `size` can be one or many numbers (sizes) or a `torch.LongStorage`. For example, `nn.CAdd(3,4,5)` is equivalent to `nn.CAdd(torch.LongStorage{3,4,5})`. If the size for a particular dimension is 1, the addition will be expanded along the entire axis.

Example:

```
mlp = nn.Sequential()
mlp:add(nn.CAdd(5, 1))

y = torch.Tensor(5, 4)
bf = torch.Tensor(5, 4)
for i = 1, 5 do bf[i] = i; end -- scale input with this

function gradUpdate(mlp, x, y, criterion, learningRate)
   local pred = mlp:forward(x)
   local err = criterion:forward(pred, y)
   local gradCriterion = criterion:backward(pred, y)
   mlp:zeroGradParameters()
   mlp:backward(x, gradCriterion)
   mlp:updateParameters(learningRate)
   return err
end

for i = 1, 10000 do
   x = torch.rand(5, 4)
   y:copy(x)
   y:add(bf)
   err = gradUpdate(mlp, x, y, nn.MSECriterion(), 0.01)
end

print(mlp:get(1).bias)
```

gives the output:

```
 1.0000
 2.0000
 3.0000
 4.0000
 5.0000
[torch.Tensor of dimension 5x1]
```

i.e. the network successfully learns the input `x` has been shifted by those bias factors to produce the output `y`.

Mul

```
module = nn.Mul()
```

Applies a *single* scaling factor to the incoming data, i.e. `y = w x`, where `w` is a scalar.

Example:

```
y = torch.Tensor(5)
mlp = nn.Sequential()
mlp:add(nn.Mul())

function gradUpdate(mlp, x, y, criterion, learningRate)
   local pred = mlp:forward(x)
   local err = criterion:forward(pred, y)
   local gradCriterion = criterion:backward(pred, y)
   mlp:zeroGradParameters()
   mlp:backward(x, gradCriterion)
   mlp:updateParameters(learningRate)
   return err
end

for i = 1, 10000 do
   x = torch.rand(5)
   y:copy(x)
   y:mul(math.pi)
```

```
   err = gradUpdate(mlp, x, y, nn.MSECriterion(), 0.01)
end

print(mlp:get(1).weight)
```

gives the output:

```
 3.1416
[torch.Tensor of dimension 1]
```

i.e. the network successfully learns the input `x` has been scaled by pi.

CMul

```
module = nn.CMul(size)
```

Applies a component-wise multiplication to the incoming data, i.e. `y_i = w_i * x_i`. Argument `size` can be one or many numbers (sizes) or a `torch.LongStorage`. For example, `nn.CMul(3,4,5)` is equivalent to `nn.CMul(torch.LongStorage{3,4,5})`. If the size for a particular dimension is 1, the multiplication will be expanded along the entire axis.

Example:

```
mlp = nn.Sequential()
mlp:add(nn.CMul(5, 1))

y = torch.Tensor(5, 4)
sc = torch.Tensor(5, 4)
for i = 1, 5 do sc[i] = i; end -- scale input with this

function gradUpdate(mlp, x, y, criterion, learningRate)
   local pred = mlp:forward(x)
   local err = criterion:forward(pred, y)
   local gradCriterion = criterion:backward(pred, y)
```

```
   mlp:zeroGradParameters()
   mlp:backward(x, gradCriterion)
   mlp:updateParameters(learningRate)
   return err
end

for i = 1, 10000 do
   x = torch.rand(5, 4)
   y:copy(x)
   y:cmul(sc)
   err = gradUpdate(mlp, x, y, nn.MSECriterion(), 0.01)
end

print(mlp:get(1).weight)
```

gives the output:

```
 1.0000
 2.0000
 3.0000
 4.0000
 5.0000
[torch.Tensor of dimension 5x1]
```

i.e. the network successfully learns the input `x` has been scaled by those scaling factors to produce the output `y`.

Max

```
module = nn.Max(dimension, nInputDim)
```

Applies a max operation over dimension `dimension`. Hence, if an `nxpxq` Tensor was given as input, and `dimension` = `2` then an `nxq` matrix would be output. When `nInputDim` is provided, inputs larger than that value will be considered batches where the actual `dimension` to apply the max operation will be dimension `dimension + 1`.

Min

```
module = nn.Min(dimension, nInputDim)
```

Applies a min operation over dimension `dimension`. Hence, if an `nxpxq` Tensor was given as input, and `dimension` = `2` then an `nxq` matrix would be output. When `nInputDim` is provided, inputs larger than that value will be considered batches where the actual `dimension` to apply the min operation will be dimension `dimension + 1`.

Mean

```
module = nn.Mean(dimension, nInputDim)
```

Applies a mean operation over dimension `dimension`. Hence, if an `nxpxq` Tensor was given as input, and `dimension` = `2` then an `nxq` matrix would be output. When `nInputDim` is provided , inputs larger than that value will be considered batches where the actual `dimension` to apply the sum operation will be dimension `dimension + 1`. This module is based on nn.Sum.

Sum

```
module = nn.Sum(dimension, nInputDim, sizeAverage, squeeze)
```

Applies a sum operation over dimension `dimension`. Hence, if an `nxpxq` Tensor was given as input, and `dimension` = `2` then an `nxq` matrix would be output. If argument `squeeze` is set to `false` then the output would be of size `nx1xq`. When `nInputDim` is provided , inputs larger than that value will be considered batches where the actual `dimension` to apply the sum operation will be dimension `dimension + 1`. Negative indexing is allowed by providing a negative value to `nInputDim`. When `sizeAverage` is provided, the sum is divided by the size of the input in this `dimension`. This is equivalent to the mean operation performed by the nn.Mean module.

Euclidean

```
module = nn.Euclidean(inputSize,outputSize)
```

Outputs the Euclidean distance of the input to `outputSize` centers, i.e. this layer has the weights `w_j`, for `j` = `1`,.., `outputSize`, where `w_j` are vectors of dimension `inputSize`.

The distance `y_j` between center `j` and input `x` is formulated as `y_j = || w_j - x ||`.

WeightedEuclidean

```
module = nn.WeightedEuclidean(inputSize,outputSize)
```

This module is similar to Euclidean, but additionally learns a separate diagonal covariance matrix across the features of the input space *for each center*.

In other words, for each of the `outputSize` centers `w_j`, there is a diagonal covariance matrices `c_j`, for `j` = `1`,.., `outputSize`, where `c_j` are stored as vectors of size `inputSize`.

The distance `y_j` between center `j` and input `x` is formulated as `y_j = || c_j * (w_j - x) ||`.

Cosine

```
module = nn.Cosine(inputSize,outputSize)
```

Outputs the cosine similarity of the input to `outputSize` centers, i.e. this layer has the weights `w_j`, for `j` = `1`,.., `outputSize`, where `w_j` are vectors of dimension `inputSize`.

The distance `y_j` between center `j` and input `x` is formulated as `y_j = (x · w_j) / ( || w_j || * || x || )`.

Kmeans

```
km = nn.Kmeans(k, dim)
```

`k` is the number of centroids and `dim` is the dimensionality of samples. The `forward` pass computes distances with respect to centroids and returns index of closest centroid. Centroids can be updated using gradient descent. Centroids can be initialized randomly or by using kmeans++ algoirthm:

```
km:initRandom(samples) -- Randomly initialize centroids from input samples.
km:initKmeansPlus(samples) -- Use Kmeans++ to initialize centroids.
```

Example showing how to use Kmeans module to do standard Kmeans clustering.

```
attempts = 10
iter = 100 -- Number of iterations
bestKm = nil
bestLoss = math.huge
learningRate = 1
for j=1, attempts do
   local km = nn.Kmeans(k, dim)
   km:initKmeansPlus(samples)
   for i=1, iter do
      km:zeroGradParameters()
      km:forward(samples) -- sets km.loss
      km:backward(samples, gradOutput) -- gradOutput is ignored

      -- Gradient Descent weight/centroids update
      km:updateParameters(learningRate)
   end

   if km.loss < bestLoss then
      bestLoss = km.loss
      bestKm = km:clone()
   end
end
```

`nn.Kmeans()` module maintains loss only for the latest forward. If you want to maintain loss over the whole dataset then you who would need do it my adding the module loss for every forward.

You can also use `nn.Kmeans()` as an auxillary layer in your network. A call to `forward` will generate an `output` containing the index of the nearest cluster for each sample in the batch. The `gradInput` generated by `updateGradInput` will be zero.

Identity

```
module = nn.Identity()
```

Creates a module that returns whatever is input to it as output. This is useful when combined with the module ParallelTable in case you do not wish to do anything to one of the input Tensors.

Example:

```
mlp = nn.Identity()
print(mlp:forward(torch.ones(5, 2)))
```

gives the output:

```
 1  1
 1  1
 1  1
 1  1
 1  1
[torch.Tensor of dimension 5x2]
```

Here is a more useful example, where one can implement a network which also computes a `Criterion` using this module:

```
pred_mlp = nn.Sequential()  -- A network that makes predictions given x.
pred_mlp:add(nn.Linear(5, 4))
pred_mlp:add(nn.Linear(4, 3))
```

```
xy_mlp = nn.ParallelTable() -- A network for predictions and for keeping the
xy_mlp:add(pred_mlp)        -- true label for comparison with a criterion
xy_mlp:add(nn.Identity())   -- by forwarding both x and y through the network.

mlp = nn.Sequential()       -- The main network that takes both x and y.
mlp:add(xy_mlp)             -- It feeds x and y to parallel networks;
cr = nn.MSECriterion()
cr_wrap = nn.CriterionTable(cr)
mlp:add(cr_wrap)            -- and then applies the criterion.

for i = 1, 100 do           -- Do a few training iterations
   x = torch.ones(5)        -- Make input features.
   y = torch.Tensor(3)
   y:copy(x:narrow(1,1,3))  -- Make output label.
   err = mlp:forward{x,y}   -- Forward both input and output.
   print(err)               -- Print error from criterion.

   mlp:zeroGradParameters() -- Do backprop...
   mlp:backward({x, y})
   mlp:updateParameters(0.05)
end
```

Copy

```
module = nn.Copy(inputType, outputType, [forceCopy, dontCast])
```

This layer copies the input to output with type casting from `inputType` to `outputType`. Unless `forceCopy` is true, when the first two arguments are the same, the input isn't copied, only transferred as the output. The default `forceCopy` is false. When `dontCast` is true, a call to `nn.Copy:type(type)` will not cast the module's `output` and `gradInput` `Tensor`s to the new type. The default is false.

Narrow

```
module = nn.Narrow(dimension, offset, length)
```

Narrow is application of narrow operation in a module. The module further supports negative `length`, `dim` and `offset` to handle inputs of unknown size.

```
> x = torch.rand(4, 5)

> x
 0.3695  0.2017  0.4485  0.4638  0.0513
 0.9222  0.1877  0.3388  0.6265  0.5659
 0.8785  0.7394  0.8265  0.9212  0.0129
 0.2290  0.7971  0.2113  0.1097  0.3166
[torch.DoubleTensor of size 4x5]

> nn.Narrow(1, 2, 3):forward(x)
 0.9222  0.1877  0.3388  0.6265  0.5659
 0.8785  0.7394  0.8265  0.9212  0.0129
 0.2290  0.7971  0.2113  0.1097  0.3166
[torch.DoubleTensor of size 3x5]

> nn.Narrow(1, 2, -1):forward(x)
 0.9222  0.1877  0.3388  0.6265  0.5659
 0.8785  0.7394  0.8265  0.9212  0.0129
 0.2290  0.7971  0.2113  0.1097  0.3166
[torch.DoubleTensor of size 3x5]

> nn.Narrow(1, 2, 2):forward(x)
 0.9222  0.1877  0.3388  0.6265  0.5659
 0.8785  0.7394  0.8265  0.9212  0.0129
[torch.DoubleTensor of size 2x5]

> nn.Narrow(1, 2, -2):forward(x)
 0.9222  0.1877  0.3388  0.6265  0.5659
 0.8785  0.7394  0.8265  0.9212  0.0129
[torch.DoubleTensor of size 2x5]

> nn.Narrow(2, 2, 3):forward(x)
 0.2017  0.4485  0.4638
 0.1877  0.3388  0.6265
```

```
 0.7394  0.8265  0.9212
 0.7971  0.2113  0.1097
[torch.DoubleTensor of size 4x3]

> nn.Narrow(2, 2, -2):forward(x)
 0.2017  0.4485  0.4638
 0.1877  0.3388  0.6265
 0.7394  0.8265  0.9212
 0.7971  0.2113  0.1097
[torch.DoubleTensor of size 4x3]
```

Replicate

```
module = nn.Replicate(nFeature [, dim, ndim])
```

This class creates an output where the input is replicated `nFeature` times along dimension `dim` (default 1). There is no memory allocation or memory copy in this module. It sets the stride along the `dim`th dimension to zero. When provided, `ndim` should specify the number of non-batch dimensions. This allows the module to replicate the same non-batch dimension `dim` for both batch and non-batch `inputs`.

```
> x = torch.linspace(1, 5, 5)
 1
 2
 3
 4
 5
[torch.DoubleTensor of dimension 5]

> m = nn.Replicate(3)
> o = m:forward(x)
 1  2  3  4  5
 1  2  3  4  5
 1  2  3  4  5
[torch.DoubleTensor of dimension 3x5]
```

```
> x:fill(13)
 13
 13
 13
 13
 13
[torch.DoubleTensor of dimension 5]

> print(o)
 13  13  13  13  13
 13  13  13  13  13
 13  13  13  13  13
[torch.DoubleTensor of dimension 3x5]
```

Reshape

```
module = nn.Reshape(dimension1, dimension2, ... [, batchMode])
```

Reshapes an `nxpxqx..` `Tensor` into a `dimension1xdimension2x...` `Tensor`, taking the elements row-wise.

The optional last argument `batchMode`, when `true` forces the first dimension of the input to be considered the batch dimension, and thus keep its size fixed. This is necessary when dealing with batch sizes of one. When `false`, it forces the entire input (including the first dimension) to be reshaped to the input size. Default `batchMode=nil`, which means that the module considers inputs with more elements than the produce of provided sizes, i.e. `dimension1xdimension2x...`, to be batches.

Example:

```
> x = torch.Tensor(4,4)
> for i = 1, 4 do
>    for j = 1, 4 do
>       x[i][j] = (i-1)*4+j
>    end
> end
> print(x)
```

```
  1   2   3   4
  5   6   7   8
  9  10  11  12
 13  14  15  16
[torch.Tensor of dimension 4x4]

> print(nn.Reshape(2,8):forward(x))

  1   2   3   4   5   6   7   8
  9  10  11  12  13  14  15  16
[torch.Tensor of dimension 2x8]

> print(nn.Reshape(8,2):forward(x))

  1   2
  3   4
  5   6
  7   8
  9  10
 11  12
 13  14
 15  16
[torch.Tensor of dimension 8x2]

> print(nn.Reshape(16):forward(x))

  1
  2
  3
  4
  5
  6
  7
  8
  9
 10
 11
 12
 13
```

```
 14
 15
 16
[torch.Tensor of dimension 16]

> y = torch.Tensor(1, 4):fill(0)
> print(y)

 0  0  0  0
 [torch.DoubleTensor of dimension 1x4]

> print(nn.Reshape(4):forward(y))

 0  0  0  0
 [torch.DoubleTensor of dimension 1x4]

> print(nn.Reshape(4, false):forward(y))

 0
 0
 0
 0
 [torch.DoubleTensor of dimension 4]
```

View

```
module = nn.View(sizes)
```

This module creates a new view of the input tensor using the `sizes` passed to the constructor. The parameter `sizes` can either be a `LongStorage` or numbers. The method `setNumInputDims()` allows to specify the expected number of dimensions of the inputs of the modules. This makes it possible to use minibatch inputs when using a size `-1` for one of the dimensions. The method `resetSize(sizes)` allows to reset the view size of the module after initialization.

Example 1:

```
> x = torch.Tensor(4, 4)
> for i = 1, 4 do
>    for j = 1, 4 do
>       x[i][j] = (i-1)*4+j
>    end
> end
> print(x)

  1   2   3   4
  5   6   7   8
  9  10  11  12
 13  14  15  16
[torch.Tensor of dimension 4x4]

> print(nn.View(2, 8):forward(x))

  1   2   3   4   5   6   7   8
  9  10  11  12  13  14  15  16
[torch.DoubleTensor of dimension 2x8]

> print(nn.View(torch.LongStorage{8,2}):forward(x))

  1   2
  3   4
  5   6
  7   8
  9  10
 11  12
 13  14
 15  16
[torch.DoubleTensor of dimension 8x2]

> print(nn.View(16):forward(x))

  1
  2
  3
  4
  5
  6
```

```
 7
 8
 9
10
11
12
13
14
15
16
[torch.DoubleTensor of dimension 16]
```

Example 2:

```
> input = torch.Tensor(2, 3)
> minibatch = torch.Tensor(5, 2, 3)
> m = nn.View(-1):setNumInputDims(2)
> print(#m:forward(input))

 6
[torch.LongStorage of size 1]

> print(#m:forward(minibatch))

 5
 6
[torch.LongStorage of size 2]
```

For collapsing non-batch dims, check out nn.Collapse.

Contiguous

```
module = nn.Contiguous()
```

Is used to make `input`, `gradOutput` or both contiguous, corresponds to `torch.contiguous` function. Only does copy and allocation if `input` or `gradOutput` is not contiguous, otherwise passes the same `Tensor`.

Select

```
module = nn.Select(dim, index)
```

Selects a dimension and index of a `nxpxqx..` `Tensor`.

Example:

```
mlp = nn.Sequential()
mlp:add(nn.Select(1, 3))

x = torch.randn(10, 5)
print(x)
print(mlp:forward(x))
```

gives the output:

```
 0.9720 -0.0836  0.0831 -0.2059 -0.0871
 0.8750 -2.0432 -0.1295 -2.3932  0.8168
 0.0369  1.1633  0.6483  1.2862  0.6596
 0.1667 -0.5704 -0.7303  0.3697 -2.2941
 0.4794  2.0636  0.3502  0.3560 -0.5500
-0.1898 -1.1547  0.1145 -1.1399  0.1711
-1.5130  1.4445  0.2356 -0.5393 -0.6222
-0.6587  0.4314  1.1916 -1.4509  1.9400
 0.2733  1.0911  0.7667  0.4002  0.1646
 0.5804 -0.5333  1.1621  1.5683 -0.1978
[torch.Tensor of dimension 10x5]

 0.0369
 1.1633
```

```
 0.6483
 1.2862
 0.6596
[torch.Tensor of dimension 5]
```

This can be used in conjunction with Concat to emulate the behavior of Parallel, or to select various parts of an input Tensor to perform operations on. Here is a fairly complicated example:

```
mlp = nn.Sequential()
c = nn.Concat(2)
for i = 1, 10 do
   local t = nn.Sequential()
   t:add(nn.Select(1, i))
   t:add(nn.Linear(3, 2))
   t:add(nn.Reshape(2, 1))
   c:add(t)
end
mlp:add(c)

pred = mlp:forward(torch.randn(10, 3))
print(pred)

for i = 1, 10000 do     -- Train for a few iterations
   x = torch.randn(10, 3)
   y = torch.ones(2, 10)
   pred = mlp:forward(x)

   criterion = nn.MSECriterion()
   err = criterion:forward(pred, y)
   gradCriterion = criterion:backward(pred, y)
   mlp:zeroGradParameters()
   mlp:backward(x, gradCriterion)
   mlp:updateParameters(0.01)
   print(err)
end
```

MaskedSelect

```
module = nn.MaskedSelect()
```

Performs a torch.MaskedSelect on a `Tensor`. The mask is supplied as a tabular argument with the input on the forward and backward passes.

Example:

```
ms = nn.MaskedSelect()
mask = torch.ByteTensor({{1, 0}, {0, 1}})
input = torch.DoubleTensor({{10, 20}, {30, 40}})
print(input)
print(mask)
out = ms:forward({input, mask})
print(out)
gradIn = ms:backward({input, mask}, out)
print(gradIn[1])
```

Gives the output:

```
10  20
30  40
[torch.DoubleTensor of size 2x2]

1  0
0  1
[torch.ByteTensor of size 2x2]

10
40
[torch.DoubleTensor of size 2]

10  0
```

```
0  40
[torch.DoubleTensor of size 2x2]
```

Index

```
module = nn.Index(dim)
```

Applies the `Tensor` index operation along the given dimension. So

```
nn.Index(dim):forward{t,i}
```

gives the same output as

```
t:index(dim, i)
```

Squeeze

```
module = nn.Squeeze([dim, numInputDims])
```

Applies the `Tensor` squeeze operation. So

```
nn.Squeeze():forward(t)
```

gives the same output as

```
t:squeeze()
```

Setting `numInputDims` allows to use this module on batches.

Unsqueeze

```
module = nn.Unsqueeze(pos [, numInputDims])
```

Insert singleton dim (i.e., dimension 1) at position `pos`. For an `input` with `dim = input:dim()`, there are `dim + 1` possible positions to insert the singleton dimension. For example, if `input` is `3` dimensional `Tensor` in size `p x q x r`, then the singleton dim can be inserted at the following `4` positions

```
pos = 1: 1 x p x q x r
pos = 2: p x 1 x q x r
pos = 3: p x q x 1 x r
pos = 4: p x q x r x 1
```

Example:

```
input = torch.Tensor(2, 4, 3) -- input: 2 x 4 x 3

-- insert at head
m = nn.Unsqueeze(1)
m:forward(input) -- output: 1 x 2 x 4 x 3

-- insert at tail
m = nn.Unsqueeze(4)
m:forward(input) -- output: 2 x 4 x 3 x 1

-- insert in between
m = nn.Unsqueeze(2)
m:forward(input) -- output: 2 x 1 x 4 x 3

-- the input size can vary across calls
input2 = torch.Tensor(3, 5, 7) -- input2: 3 x 5 x 7
m:forward(input2) -- output: 3 x 1 x 5 x 7
```

Indicate the expected input feature map dimension by specifying `numInputDims`. This allows the module to work with mini-batch. Example:

```
b = 5 -- batch size 5
input = torch.Tensor(b, 2, 4, 3) -- input: b x 2 x 4 x 3
numInputDims = 3 -- input feature map should be the last 3 dims

m = nn.Unsqueeze(4, numInputDims)
m:forward(input) -- output: b x 2 x 4 x 3 x 1

m = nn.Unsqueeze(2):setNumInputDims(numInputDims)
m:forward(input) -- output: b x 2 x 1 x 4 x 3
```

Transpose

```
module = nn.Transpose({dim1, dim2} [, {dim3, dim4}, ...])
```

Swaps dimension `dim1` with `dim2`, then `dim3` with `dim4`, and so on. So

```
nn.Transpose({dim1, dim2}, {dim3, dim4}):forward(t)
```

gives the same output as

```
t:transpose(dim1, dim2)
t:transpose(dim3, dim4)
```

The method `setNumInputDims()` allows to specify the expected number of dimensions of the inputs of the modules. This makes it possible to use minibatch inputs. Example:

```
b = 5 -- batch size 5
input = torch.Tensor(b, 2, 4, 3) -- input: b x 2 x 4 x 3

m = nn.Transpose({1,3})
m:forward(input) -- output: 4 x 2 x b x 3 x 1

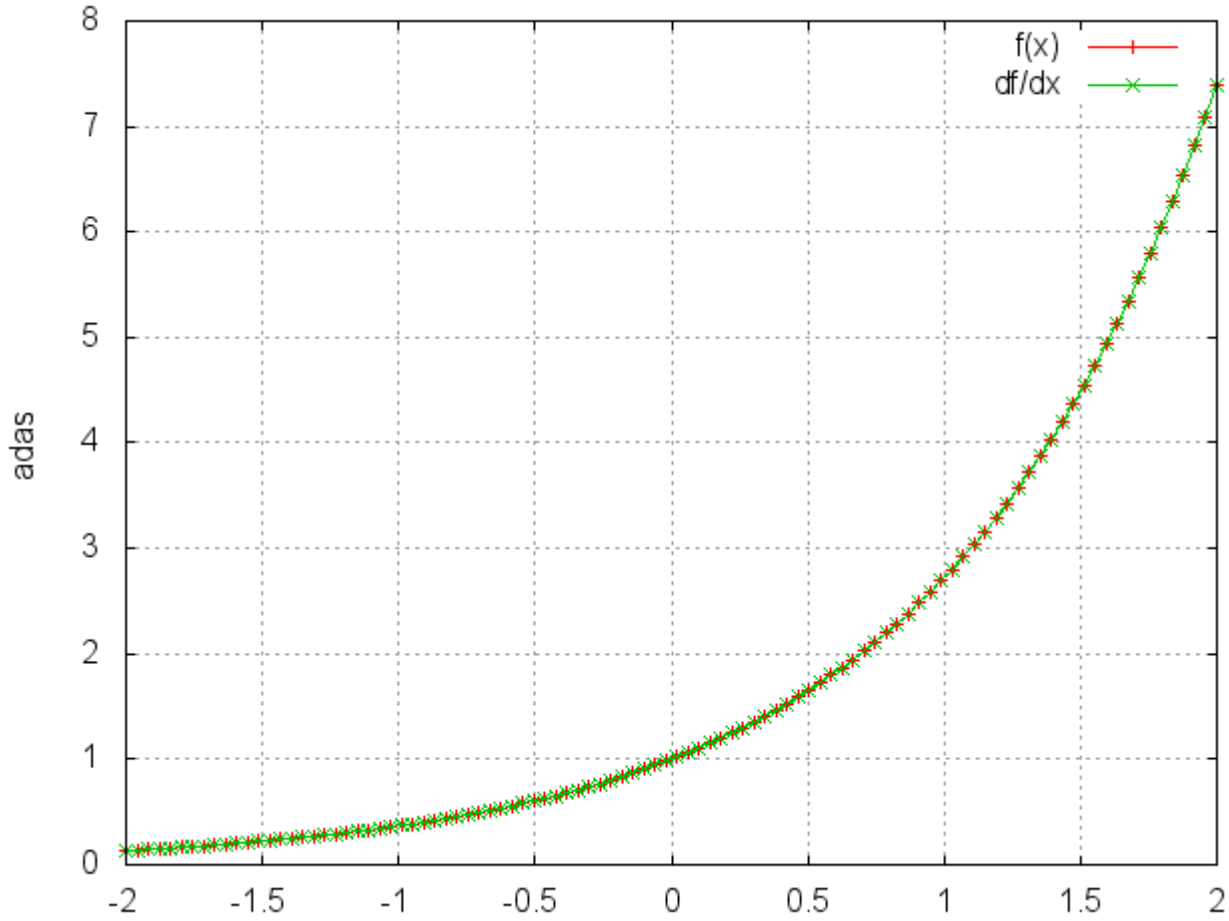
numInputDims = 3 -- input feature map should be the last 3 dims
m = nn.Transpose({1,3}):setNumInputDims(numInputDims)
m:forward(input) -- output: b x 3 x 4 x 2
```

Exp

```
module = nn.Exp()
```

Applies the `exp` function element-wise to the input `Tensor`, thus outputting a `Tensor` of the same dimension.

```
ii = torch.linspace(-2, 2)
m = nn.Exp()
oo = m:forward(ii)
go = torch.ones(100)
gi = m:backward(ii,go)
gnuplot.plot({'f(x)', ii, oo, '+-'}, {'df/dx', ii, gi, '+-'})
gnuplot.grid(true)
```



Log

```
module = nn.Log()
```

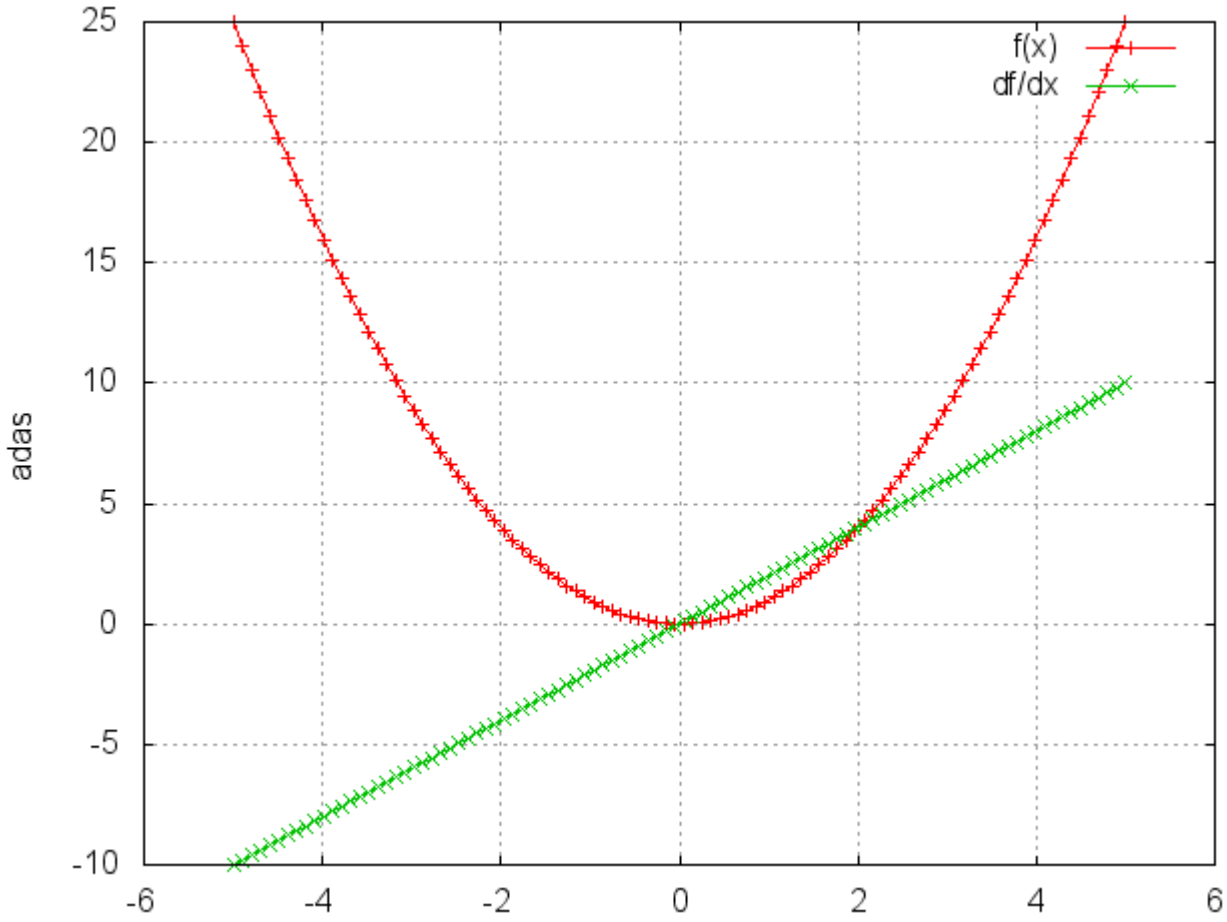
Applies the `log` function element-wise to the input `Tensor`, thus outputting a Tensor of the same dimension.

Square

```
module = nn.Square()
```

Takes the square of each element.

```
ii = torch.linspace(-5, 5)
m = nn.Square()
oo = m:forward(ii)
go = torch.ones(100)
gi = m:backward(ii, go)
gnuplot.plot({'f(x)', ii, oo, '+-'}, {'df/dx', ii, gi, '+-'})
gnuplot.grid(true)
```

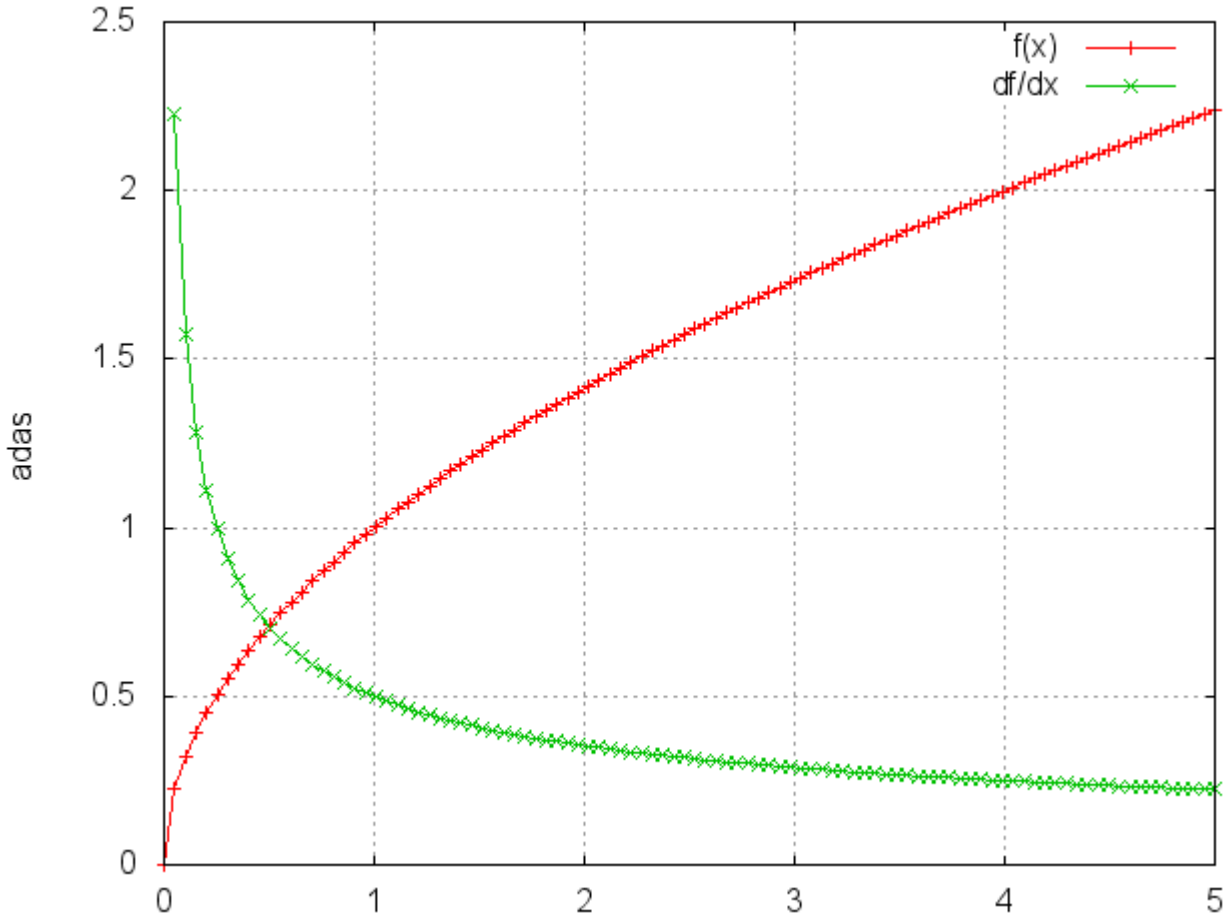


Sqrt

```
module = nn.Sqrt()
```

Takes the square root of each element.

```
ii = torch.linspace(0, 5)
m = nn.Sqrt()
oo = m:forward(ii)
go = torch.ones(100)
gi = m:backward(ii, go)
gnuplot.plot({'f(x)', ii, oo, '+-'}, {'df/dx', ii, gi, '+-'})
gnuplot.grid(true)
```



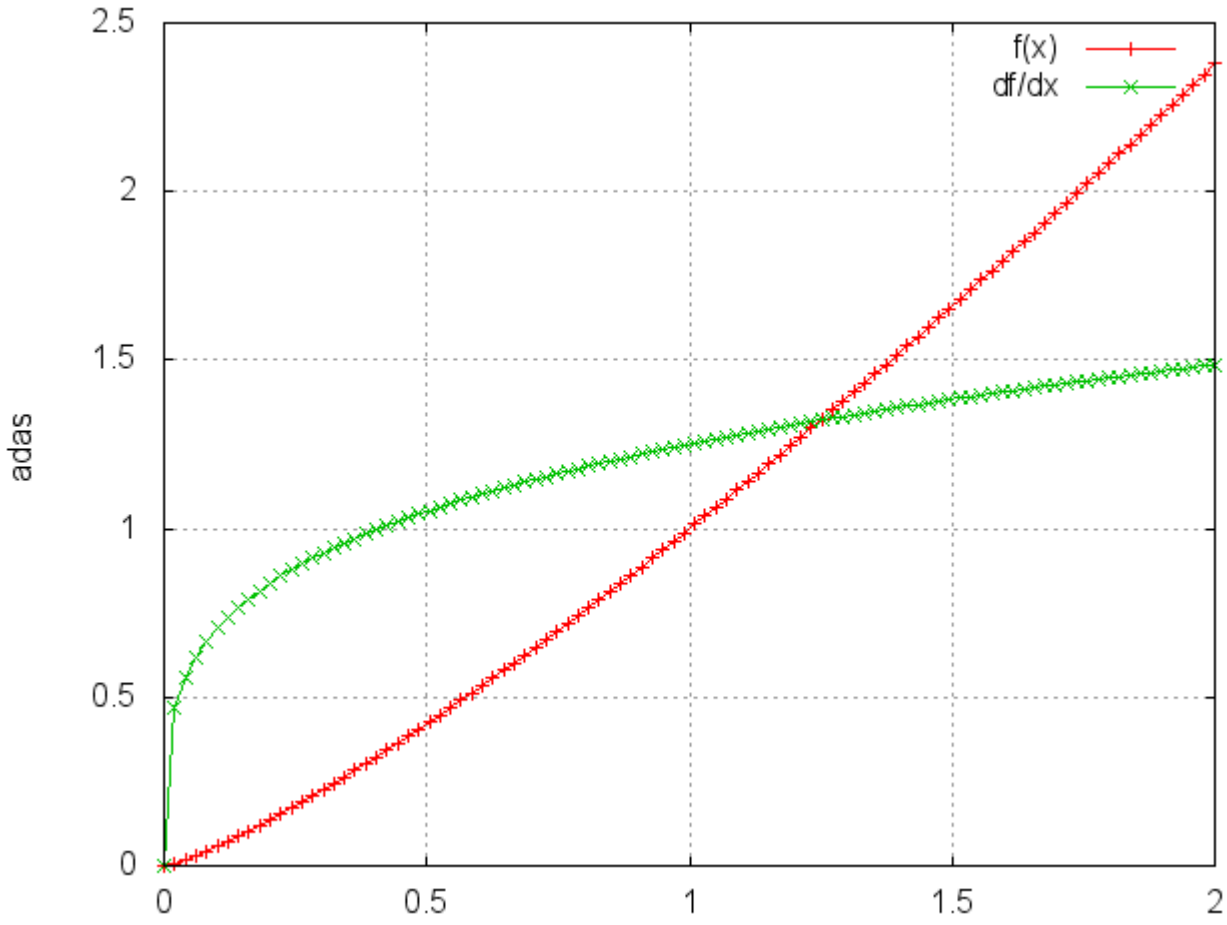
Power

```
module = nn.Power(p)
```

Raises each element to its `p`-th power.

```
ii = torch.linspace(0, 2)
m = nn.Power(1.25)
oo = m:forward(ii)
go = torch.ones(100)
```

```
gi = m:backward(ii, go)
gnuplot.plot({'f(x)', ii, oo, '+-'}, {'df/dx', ii, gi, '+-'})
gnuplot.grid(true)
```



Clamp

```
module = nn.Clamp(min_value, max_value)
```

Clamps all elements into the range `[min_value, max_value]`. Output is identical to input in the range, otherwise elements less than `min_value` (or greater than `max_value`) are saturated to `min_value` (or `max_value`).

```
A = torch.randn(2, 5)
m = nn.Clamp(-0.1, 0.5)
B = m:forward(A)

print(A)  -- input
-1.1321  0.0227 -0.4672  0.6519 -0.5380
 0.9061 -1.0858  0.3697 -0.8120 -1.6759
[torch.DoubleTensor of size 3x5]

print(B)  -- output
-0.1000  0.0227 -0.1000  0.5000 -0.1000
 0.5000 -0.1000  0.3697 -0.1000 -0.1000
[torch.DoubleTensor of size 3x5]
```

Normalize

```
module = nn.Normalize(p, [eps])
```

Normalizes the input `Tensor` to have unit `L_p` norm. The smoothing parameter `eps` prevents division by zero when the input contains all zero elements (default = `1e-10`).

Input can be 1D or 2D (in which case it's considered as in batch mode)

```
A = torch.randn(3, 5)
m = nn.Normalize(2)
B = m:forward(A) -- B is also 3 x 5
-- take the L2 norm over the second axis:
print(torch.norm(B, 2, 2)) -- norms is [1, 1, 1]
```

`Normalize` has a specialized implementation for the `inf` norm, which corresponds to the maximum norm.

```
A = torch.randn(3,5)
m = nn.Normalize(math.huge) -- uses maximum/inf norm
B = m:forward(A)
maxA = torch.abs(A):max(2)
print(A,B,maxA)
```

MM

```
module = nn.MM(transA, transB)
```

Performs multiplications on one or more pairs of matrices. If `transA` is set to true, the first matrix is transposed before multiplication. If `transB` is set to true, the second matrix is transposed before multiplication. By default, the matrices do not get transposed.

The module also accepts 3D inputs which are interpreted as batches of matrices. When using batches, the first input matrix should be of size `b x m x n` and the second input matrix should be of size `b x n x p` (assuming `transA` and `transB` are not set). If `transA` or `transB` is set, transpose takes place between the second and the third dimensions for the corresponding matrix.

```
model = nn.MM()
A = torch.randn(b, m, n)
B = torch.randn(b, n, p)
C = model:forward({A, B})  -- C will be of size `b x m x p`

model = nn.MM(true, false)
A = torch.randn(b, n, m)
B = torch.randn(b, n, p)
C = model:forward({A, B})  -- C will be of size `b x m x p`
```

BatchNormalization

```
module = nn.BatchNormalization(N [, eps] [, momentum] [,affine])
```

where `N` is the dimensionality of input `eps` is a small value added to the standard-deviation to avoid divide-by-zero. Defaults to `1e-5`. `affine` is a boolean. When set to false, the learnable affine transform is disabled. Defaults to true

During training, this layer keeps a running estimate of its computed mean and std. The running sum is kept with a default momentum of 0.1 (unless over-ridden) During evaluation, this running mean/std is used for normalization.

Implements Batch Normalization as described in the paper: "Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift" by Sergey Ioffe, Christian Szegedy.

The operation implemented is:

```
              x - mean(x)
y =  ----------------------------- * gamma + beta
      standard-deviation(x) + eps
```

where the mean and standard-deviation are calculated per-dimension over the mini-batches and where gamma and beta are learnable parameter vectors of size `N` (where `N` is the input size). The learning of gamma and beta is optional. The module only accepts 2D inputs.

```
-- with learnable parameters
model = nn.BatchNormalization(m)
A = torch.randn(b, m)
C = model:forward(A)  -- C will be of size `b x m`

-- without learnable parameters
model = nn.BatchNormalization(m, nil, nil, false)
A = torch.randn(b, m)
C = model:forward(A)  -- C will be of size `b x m`
```

PixelShuffle

`module = nn.PixelShuffle(r)`

Rearranges elements in a tensor of shape `[C*r*r, H, W]` to a tensor of shape `[C, H*r, W*r]`. This is useful for implementing efficient sub-pixel convolution with a stride of `1/r` (see Shi et. al). Below we show how the `PixelShuffle` module can be used to learn upscaling filters to transform a low-resolution input to a high resolution one, with a 3x upscale factor. This is useful for tasks such as super-resolution, see "Real-Time Single Image and Video Super-Resolution Using an Efficient Sub-Pixel Convolutional Neural Network" - Shi et al. for further details.

```
upscaleFactor = 3
inputChannels = 1

model = nn.Sequential()
model:add(nn.SpatialConvolution(inputChannels, 64, 5, 5, 1, 1, 2, 2))
model:add(nn.ReLU())

model:add(nn.SpatialConvolution(64, 32, 3, 3, 1, 1, 1, 1))
model:add(nn.ReLU())

model:add(nn.SpatialConvolution(32, inputChannels * upscaleFactor * upscaleFactor, 3, 3, 1, 1, 1, 1))
model:add(nn.PixelShuffle(upscaleFactor))

input = torch.Tensor(1, 192, 256);
out = model:forward(input)
out:size()
   1
 576
 768
[torch.LongStorage of size 3]
```

Padding

```
module = nn.Padding(dim, pad [, nInputDim, value, index])
```

This module adds `pad` units of padding to dimension `dim` of the input. If `pad` is negative, padding is added to the left, otherwise, it is added to the right of the dimension. When `nInputDim` is provided, inputs larger than that value will be considered batches where the

actual `dim` to be padded will be dimension `dim + 1`. When `value` is provide, the padding will be filled with that `value`. The default `value` is zero. When `index` is provided, padding will be added at that offset from the left or right, depending on the sign of `pad`.

Example 1:

```
module = nn.Padding(1, 2, 1, -1) --pad right x2
module:forward(torch.randn(3)) --non-batch input
 0.2008
 0.4848
-1.0783
-1.0000
-1.0000
[torch.DoubleTensor of dimension 5]
```

Example 2:

```
module = nn.Padding(1, -2, 1, -1) --pad left x2
module:forward(torch.randn(2, 3)) --batch input
-1.0000 -1.0000  1.0203  0.2704 -1.6164
-1.0000 -1.0000 -0.2219 -0.6529 -1.9218
[torch.DoubleTensor of dimension 2x5]
```

Example 3:

```
module = nn.Padding(1, -2, 1, -1, 2) --pad left x2, offset to index 2
module:forward(torch.randn(2, 3)) --batch input
 1.0203 -1.0000 -1.0000  0.2704 -1.6164
-0.6529 -1.0000 -1.0000 -0.2219 -1.9218
[torch.DoubleTensor of dimension 2x5]
```

L1Penalty

```
```

```
penalty = nn.L1Penalty(L1weight, sizeAverage)
```

L1Penalty is an inline module that in its forward propagation copies the input Tensor directly to the output, and computes an L1 loss of the latent state (input) and stores it in the module's `loss` field. During backward propagation: `gradInput = gradOutput + gradLoss`.

This module can be used in autoencoder architectures to apply L1 losses to internal latent state without having to use Identity and parallel containers to carry the internal code to an output criterion.

Example (sparse autoencoder, note: decoder should be normalized):

```
encoder = nn.Sequential()
encoder:add(nn.Linear(3, 128))
encoder:add(nn.Threshold())
decoder = nn.Linear(128, 3)

autoencoder = nn.Sequential()
autoencoder:add(encoder)
autoencoder:add(nn.L1Penalty(l1weight))
autoencoder:add(decoder)

criterion = nn.MSECriterion()  -- To measure reconstruction error
-- ...
```

GradientReversal

```
module = nn.GradientReversal([lambda = 1])
```

This module preserves the input, but takes the gradient from the subsequent layer, multiplies it by `-lambda` and passes it to the preceding layer. This can be used to maximise an objective function whilst using gradient descent, as described in "Domain-Adversarial Training of Neural Networks" (http://arxiv.org/abs/1505.07818).

One can also call:

```
module:setLambda(lambda)
```

to set the hyper-parameter `lambda` dynamically during training.

GPU

```
gpu = nn.GPU(module, device, [outdevice])
require 'cunn'
gpu:cuda()
```

Decorates an encapsulated `module` so that it can be executed on a specific GPU `device`. The decorated module's `parameters` are thus hosted on the specified GPU `device`. All operations on the `gpu` module are executed on that device. Calls to `forward`/`backward` will transfer arguments `input` and `gradOutput` to the specified `device`, which are then fed as arguments to the decorated `module`. Returned `output` is located on the specified `outdevice` (defaults to `device`). Returned `gradInput` is allocated on the same device as the `input`.

When serialized/deserialized, the `gpu` module will be run on the same `device` that it was serialized with. To prevent this from happening, the module can be converted to float/double before serialization:

```
gpu:float()
gpustr = torch.serialize(gpu)
```

The module is located in the **nn** package instead of **cunn** as this allows it to be used in CPU-only environments, which are common for production models.

The module supports nested table `input` and `gradOutput` tensors originating from multiple devices. Each nested tensor in the returned `gradInput` will be transferred to the device its commensurate tensor in the `input`.

The intended use-case is not for model-parallelism where the models are executed in parallel on multiple devices, but for sequential models where a single GPU doesn't have enough memory.

Example using 4 GPUs:

```
mlp = nn.Sequential()
   :add(nn.GPU(nn.Linear(10000,10000), 1))
   :add(nn.GPU(nn.Linear(10000,10000), 2))
   :add(nn.GPU(nn.Linear(10000,10000), 3))
   :add(nn.GPU(nn.Linear(10000,10000), 4, cutorch.getDevice()))
```

Note how the last `GPU` instance will return an `output` tensor on the same device as the current device (`cutorch.getDevice`).

TemporalDynamicKMaxPooling

```
module = nn.TemporalDynamicKMaxPooling(minK, [factor])
```

Selects the highest `k` values for each feature in the feature map sequence provided. The input sequence is composed of `nInputFrame` frames (i.e. `nInputFrame` is sequence length). The `input` tensor in `forward(input)` is expected to be a 2D tensor (`nInputFrame x inputFrameSize`) or a 3D tensor (`nBatchFrame x nInputFrame x inputFrameSize`), where `inputFrameSize` is the number of features across the sequence.

If `factor` is not provided, `k = minK`, else the value of k is calculated with:

```
k = math.max(minK, math.ceil(factor*nInputFrame)))
```

Constant

```
module = nn.Constant(value, nInputDim)
```

This module outputs a constant value given an input. If `nInputDim` is specified, it uses the input to determine the size of the batch. The `value` is then replicated over the batch. Otherwise, the `value` Tensor is output as is. During `backward`, the returned `gradInput` is a zero Tensor of the same size as the `input`. This module has no trainable parameters.

You can use this with nn.ConcatTable() to append constant inputs to an input :

```
nn.ConcatTable():add(nn.Constant(v)):add(nn.Identity())
```

This is useful when you want to output a value that is independent of the input to the neural network.

WhiteNoise

```
module = nn.WhiteNoise([mean, stdev])
```

This module adds isotropic Gaussian noise to the `input`. This can be useful for training Denoising Autoencoders. Takes `mean` and `stdev` of the normal distribution as constructor arguments. Default values for mean and standard deviation are 0 and 0.1 respectively. With `module:training()`, Gaussian noise is added during `forward`. During `backward` gradients are passed as is. With `module:evaluate()` the `mean` is added to the input.

OneHot

```
module = nn.OneHot(outputSize)
```

Transforms a tensor of `input` indices having integer values between 1 and `outputSize` into a tensor of one-hot vectors of size `outputSize`.

Forward an index to get a one-hot vector :

```
> module = nn.OneHot(5) -- 5 classes
> module:forward(torch.LongTensor{3})
 0  0  1  0  0
[torch.DoubleTensor of size 1x5]
```

Forward a batch of 3 indices. Notice that these need not be stored as `torch.LongTensor` :

```
> module:forward(torch.Tensor{3,2,1})
 0  0  1  0  0
 0  1  0  0  0
 1  0  0  0  0
[torch.DoubleTensor of size 3x5]
```

Forward batch of `2 x 3` indices :

```
oh:forward(torch.Tensor{{3,2,1},{1,2,3}})
(1,.,.) =
  0  0  1  0  0
  0  1  0  0  0
  1  0  0  0  0

(2,.,.) =
  1  0  0  0  0
  0  1  0  0  0
  0  0  1  0  0
[torch.DoubleTensor of size 2x3x5]
```

PrintSize

```
module = nn.PrintSize(name)
```

This module is useful for debugging complicated module composites. It prints the size of the `input` and `gradOutput` during `forward` and `backward` propagation respectively. The `name` is a string used to identify the module along side the printed size.

ZeroGrad

```
module = nn.ZeroGrad()
input = torch.Tensor{1,2}
gradOutput = torch.Tensor{3,4}
print(module:forward(input))
 1
 2
[torch.DoubleTensor of size 2]

print(module:backward(input, gradOutput))
 0
 0
[torch.DoubleTensor of size 2]
```

The module zeros the `gradInput` but forwards the `input` as-is.

Collapse

```
module = nn.Collapse(nInputDim)
```

This module is the equivalent of:

```
view = nn.View(-1)
view:setNumInputDim(nInputDim)
```

It collapses all non-batch dimensions. This is useful for converting a spatial feature map to the single dimension required by a dense hidden layer like Linear.

Convert

```
module = nn.Convert([inputShape, outputShape])
```

Module to convert between different data formats. For example, we can flatten images by using :

```
module = nn.Convert('bchw', 'bf')
```

or equivalently

```
module = nn.Convert('chw', 'f')
```

Lets try it with an input:

```
print(module:forward(torch.randn(3,2,3,1)))
 0.5692 -0.0190  0.5243  0.7530  0.4230  1.2483
-0.9142  0.6013  0.5608 -1.0417 -1.4014  1.0177
-1.5207 -0.1641 -0.4166  1.4810 -1.1725 -1.0037
[torch.DoubleTensor of size 3x6]
```

You could also try:

```
module = nn.Convert('chw', 'hwc')
input = torch.randn(1,2,3,2)
input:select(2,1):fill(1)
input:select(2,2):fill(2)
print(input)
(1,1,.,.) =
  1  1
  1  1
  1  1
(1,2,.,.) =
  2  2
  2  2
  2  2
[torch.DoubleTensor of size 1x2x3x2]
print(module:forward(input))
(1,1,.,.) =
```

```
  1   2
  1   2

(1,2,.,.) =
  1   2
  1   2

(1,3,.,.) =
  1   2
  1   2
[torch.DoubleTensor of size 1x3x2x2]
```

Furthermore, it automatically converts the `input` to have the same type as `self.output` (i.e. the type of the module). So you can also just use is for automatic input type converions:

```
module = nn.Convert()
print(module.output) -- type of module
[torch.DoubleTensor with no dimension]
input = torch.FloatTensor{1,2,3}
print(module:forward(input))
 1
 2
 3
[torch.DoubleTensor of size 3]
```